(AFRL), is a pioneering example of a low-cost, stealthy, autonomous “loyal wingman” UAV. Designed under the Low-cost Attritable Aircraft Technology (LCAAT) program, the Valkyrie demonstrates how modern drones can operate cooperatively with manned fighter aircraft, enhancing mission capability while reducing risk to human pilots (Wang, et al. 2020).

The UAV features a blended-wing airframe optimized for radar stealth and aerodynamic efficiency, with a cruise speed exceeding 0.85 Mach and an operational ceiling of 45,000 feet. Its modular payload bay supports a wide range of mission’s surveillance, electronic warfare, and precision strike, making it a flexible combat companion for aircraft such as the F-35 Lightning II and F-22 Raptor.

Equipped with an AI-based flight control system, the Valkyrie can execute autonomous mission planning, formation flying, and threat response while maintaining secure data links through encrypted mesh networks. Its open-architecture design allows integration of different payloads and systems, ensuring future adaptability for multi-domain operations.

Strategically, the XQ-58A Valkyrie represents a critical step toward autonomous combat collaboration, enabling manned–unmanned teaming (MUM-T) in contested airspaces. Its combination of low cost, stealth, and AI-driven autonomy makes it a foundational model for next-generation aerial warfare, influencing programs like Skyborg and Next Generation Air Dominance (NGAD).

Figure 5.3 showcases the XQ-58A Valkyrie, an advanced autonomous stealth combat UAV developed by Kratos Defense and the US Air Force Research Laboratory (AFRL) under the Low-cost Attritable Aircraft